SPEAKER SCRIPT

Is South Carolina Heating Up?

15-minute talk · 16 slides · about 1,900 words (≈ 135 words per minute plus pauses)

How to use this script

- The same text is in the Notes of every slide in SC_Climate_Presentation.pptx.
- In PowerPoint: Slide Show → tick "Use Presenter View" (or press Alt+F5). You will see your notes and a timer; the audience sees only the slides.
- Words in [square brackets and red] are stage directions – point at the chart, pause – don't read them out.
- Don't read word for word. Read it out loud 2–3 times beforehand, then use it as a safety net.
- Before you present: replace [Your Name], [Course name], [Date] and [GitHub repository link] on slides 1 and 16.

Timing plan

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Script, slide by slide

Slide 1 – Title 0:00–0:45

Good *[morning/afternoon]* everyone. My name is *[Your Name]*, and my project asks a simple question: is South Carolina heating up?

The coloured band at the bottom of this slide is my whole project in one picture. Each stripe is one year, from 1966 on the left to 2025 on the right. Blue years were cooler than normal and red years were warmer. You can see the colours change as you go from left to right.

In the next fifteen minutes I'll show you where I got the data, the three angles I used to analyse it, and what I found – including one result that surprised me.

Slide 2 – Why this topic 0:45–1:45

Why South Carolina? The assignment pointed us to NOAA's past-weather page for Charleston, and Charleston is a great example of why weather matters.

[Point to the cards] One – floods: Charleston floods at high tide, and in October 2015 the state had a so-called thousand-year flood. Two – heat: summers are hot and humid, and hot nights affect people's health and their electricity bills. Three – farms: agriculture is a big part of the economy, so freeze dates and the growing season really matter.

We usually hear about climate change as one global number, but people feel the weather where they live. So my question was: what has actually changed in South Carolina's weather over the last 60 years – where, when, and how do people feel it? And I wanted to answer it with public data that anyone can check, using real statistics instead of just eyeballing a chart.

Slide 3 – The data 1:45–2:45

All my data comes from NOAA's Global Historical Climatology Network – Daily. It's the same database that powers the past-weather page in the assignment. Every station has one CSV file you can download straight from a URL – no account needed.

I filtered NOAA's station list down to South Carolina, downloaded 18 candidate stations, and kept the 10 that have a complete record – at least 330 days of data – in almost every year from 1966 to 2025.

[Point to the map] They cover the three regions people in the state talk about: the Coast in blue, the Midlands in orange and the Upstate in green. Charleston has two stations, the airport and downtown, and that turned out to be really useful later.

In total that's 217,782 days of daily highs, lows and rainfall. My download script saves a trimmed copy into the repository, so the project runs even without internet.

Slide 4 – Cleaning & quality check 2:45–4:00

Raw weather data is messy, so before any analysis I cleaned it. *[Point to the list]* First, NOAA stores temperatures in tenths of a degree Celsius and rain in tenths of a millimetre, so I converted everything to Fahrenheit and inches. Second, every value has a quality flag, and I removed the ones that failed NOAA's checks. Third, I only use years with at least 330 days of data, so a missing summer can't make a year look cool.

Then I added one check of my own. Stations in the same state go up and down together, so I compared each station with the average of the other nine. *[Point to the grey lines]* Each grey line is a healthy station – they all stay close to zero. *[Point to the red line]* The red line is Sumter. Around 2010 to 2015 its thermometer suddenly reads up to 5 degrees colder than its neighbours. Columbia, about 40 miles away, shows nothing like it, so this is a broken or moved instrument, not real weather.

So I left Sumter out of all the statewide temperature averages. Without this check, some of my answers would have been wrong.

Slide 5 – The three angles 4:00–5:30

Now the core of the project: the three angles. A single number like "the average temperature went up" can't tell a city where to act, a doctor when people are at risk, or a farmer when to plant. So each angle answers the question the previous one leaves open.

Angle one is WHERE: which city has warmed the most? That's the example question from the assignment, and it tells us whether warming is a local thing or statewide.

Angle two is WHEN: is it warming more during the day or at night, and in which season? The yearly average hides that, and the answer points to causes – like the heat that cities themselves create.

Angle three is WHAT PEOPLE FEEL: extreme days. Nobody feels an average. People feel a 95-degree afternoon, a night that never cools down, a frost or a flood. This angle also lets me test rainfall, which is a second variable.

[Point to the bottom line] So we go from the average, to its timing, to the extremes – space, time and impact. All three use the same cleaned data and the same statistics: a straight-line trend with a p-value, plus a simple comparison of 1966–1995 against 1996–2025.

Slide 6 – Angle 1 – statewide 5:30–6:30

Let's start with the big picture. Each bar is one year compared with the 1966 to 1995 average – blue is cooler, red is warmer. The black line is a 10-year average.

Before about 1985 most years are blue; since 2010 nearly every year is red. The trend is plus 0.49 degrees Fahrenheit per decade, with a p-value below 0.001 – so it's very unlikely to be random noise. Over 60 years that adds up to about 2.9 degrees.

The warmest single year was actually 1990, but 8 of the 10 warmest years came after 2010. So yes – South Carolina is heating up. Now: where?

Slide 7 – Angle 1 – which city 6:30–7:45

This chart answers Angle one directly. Each bar is a city's warming rate, the colour shows its region, and the thin lines are 95 percent confidence intervals.

Columbia has warmed the fastest, at 0.58 degrees per decade – that's about 3.5 degrees since 1966. But look at the error bars: Greenville-Spartanburg, Beaufort and Charleston Airport are within 0.03 of Columbia, and their intervals overlap almost completely. So the honest answer is: Columbia comes first, but the top four are statistically tied.

The more important finding is that every single trusted station warmed, all with p below 0.001, and the three regions are very close: Coast 0.53, Midlands 0.50, Upstate 0.45. Sumter, in grey at the bottom, is the broken station from before.

Slide 8 – Angle 1 – warming stripes 7:45–8:30

Here's the same story without any statistics. Each row is a city, each column is a year. Red means warmer than that city's own normal, blue means cooler.

The left side is mostly blue and the right side is mostly red – at every station, and at the same time. That tells us this is a statewide change, not just one city's story.

[Point to the bottom row] You can also spot Sumter's broken thermometer: it's the only row that turns dark blue around 2013.

Slide 9 – Angle 2 – day vs night 8:30–9:45

Angle two: when is it warming? Here I split each day into the afternoon high, in orange, and the night-time low, in purple.

On average, nights warm at 0.55 degrees per decade and days at 0.43 – so nights are warming faster. But look at the top two rows: downtown Charleston and Columbia. Downtown

Charleston's nights warmed more than three times faster than its days. These are the two most built-up locations in my data.

That's the urban heat island effect: concrete, asphalt and buildings soak up heat during the day and release it at night, so cities stay warm after sunset. Smaller places like Anderson and Rock Hill actually warm a little more by day. So part of what city residents feel is extra warming caused by the city itself, on top of the regional trend.

Slide 10 – Angle 2 – seasons 9:45–10:35

Next: which season. Winter is warming fastest – about 0.7 degrees per decade, which is 1.7 times faster than the other three seasons. And in spring, summer and fall, the purple night bars are again taller than the orange day bars.

One caution: winter weather varies a lot from year to year, so the error bars are wide. The winter result is significant, but the exact number 1.7 is less certain.

Put together, Angle two says South Carolina is warming mostly in winter and at night – it's not just hotter summer afternoons.

Slide 11 – Angle 3 – hot days & warm nights 10:35–11:45

Angle three: what people actually feel. On the left are hot days, with a high of 95 or more; on the right are warm nights that never drop below 75. The open circle is 1966 to 1995, and the filled circle is 1996 to 2025.

Warm nights changed the most: statewide they're up 47 percent. Downtown Charleston now averages 84 nights a year that never cool below 75 – up from 59. That's almost an extra month of nights with no relief, which matters for health and for electricity bills. Columbia went from about 5 to 19.

Hot days rose too, about 21 percent statewide, but that change is only just significant. And interestingly, downtown Charleston has fewer hot afternoons than before – the sea breeze keeps the days in check, but the nights don't cool off.

Slide 12 – Angle 3 – freezes 11:45–12:30

At the cold end: freezing nights are down about 18 percent, roughly 3 fewer per decade. And the freeze-free season – basically the growing season – is getting longer by about 3.4 days per decade. That's around three weeks longer than in the 1960s.

That's good news for some crops, but it also means a longer season for pests, pollen and mosquitoes.

Slide 13 – Angle 3 – rain 12:30–13:20

Now the surprise. I expected rain to get more extreme too. This chart shows the number of days with 2 or more inches of rain.

The biggest spikes are single storms – Hurricane David in 1979, the October 2015 flood, Hurricane Matthew in 2016, and Debby and Helene in 2024 – and big storm years happen in every decade. The trend is basically zero, with a p-value of 0.56, so it's not significant. The 10-year average has gone up since about 2010, so it's worth watching, but with this data I can't claim heavy rain has increased.

I think that's an important result: not every climate claim is equally supported by local data.

Slide 14 – Scorecard 13:20–14:00

Here's everything from Angle three on one scorecard. Coloured bars have a significant 60-year trend; grey bars don't.

Every temperature extreme moved: more hot days, many more warm nights, fewer freezes and a longer growing season. Every rain measure is grey – total rain, heavy-rain days, the wettest day and dry spells show no clear change.

Slide 15 – Conclusion 14:00–14:45

To wrap up.

Where: South Carolina has warmed about 3 degrees since 1966, at every trusted station. Columbia is fastest, but it's statistically tied with three others.

When: the warming is strongest in winter and at night, and the night-time warming is strongest in the cities.

What people feel: many more warm nights, fewer freezes and a growing season about three weeks longer – but no clear change in heavy rain yet.

There are limitations: I used raw station data with a simple quality check, cities have grown around some of the stations, and a straight-line trend is a simple model. But the picture is consistent across all three angles. And one lesson for data science: always check your data – one broken thermometer in Sumter would have changed the answer.

Slide 16 – Thank you 14:45–15:00

Thank you for listening. All the code, the data and the report are in my GitHub repository, and running one command – python run_all.py – rebuilds every chart you saw today.

I'm happy to take any questions.

Likely questions – and good answers

Q: Why did you start in 1966 and not earlier?

A: 1966–2025 is the longest period where all 10 stations have a complete record. Starting earlier would have meant dropping Greenville-Spartanburg and Beaufort. It also gives two equal 30-year periods, the same length NOAA uses for climate normals.

Q: Isn't removing Sumter cherry-picking?

A: No – the rule was set before looking at the trends: any station that drifts more than 3 °F from its neighbours is flagged. All other stations stayed within about 2 °F; Sumter drifted 5 °F. Sumter is still shown on every chart, in grey, and its rain data is still used.

Q: How do you know the warming is real and not just the cities growing?

A: City growth explains part of the night-time warming at downtown Charleston and Columbia. But smaller, more rural stations like Anderson, Rock Hill and Florence also warmed significantly, and the warming shows up everywhere at the same time – that points to a regional change, with extra urban warming on top.

Q: What does the p-value mean here?

A: It is the chance of seeing a trend this strong if there were really no trend and the years were just random noise. For the statewide warming it is far below 0.001, so random noise is very unlikely. For heavy rain it is 0.56, so a trend that size could easily be chance.

Q: Why a straight line? Climate isn't linear.

A: A straight line is the simplest summary of the overall direction, and it is easy to explain. I also compared two 30-year averages, which doesn't assume a straight line, and both methods give the same picture. A curved model would be a good next step.

Q: Why 95 °F and 75 °F as thresholds?

A: They're common, easy-to-understand thresholds for heat stress in the Southeast: a 95 °F afternoon is a heat-advisory type day, and a night that stays above 75 °F gives the body and air-conditioners no break. The method works for any threshold – it's one line in config.py.

Q: Why does rain show no trend when the news says storms are getting worse?

A: Rain is much noisier than temperature – one hurricane can double a year's count. With 60 years and 10 stations the change isn't distinguishable from chance yet. It doesn't prove there's no change; it means this data can't show one. The 10-year average has been rising since about 2010, so it's worth re-checking in a few years.

Q: How can someone repeat your analysis?

A: Clone the GitHub repository, run pip install -r requirements.txt, then python run_all.py. It rebuilds every number and chart in about 20 seconds from the data in the repository; add --download to fetch fresh data from NOAA.

Recording the YouTube video

Option A – record inside PowerPoint (easiest)

1. Open SC_Climate_Presentation.pptx → Slide Show tab → Record (Record from Beginning).
2. Turn on the microphone (and camera if you want your face in the corner). Your notes are shown at the top of the recording window.
3. Click the red record button, present normally, and press the right arrow to move on. Stop after the last slide.
4. File → Export → Create a Video → Full HD (1080p) → Use Recorded Timings and Narrations → Create Video. Save as an .mp4.

Option B – screen recording

- Windows: press Win+Alt+R (Xbox Game Bar) or use the Snipping Tool's video mode while the slide show is running. Zoom or OBS Studio also work.

Upload to YouTube

1. Go to youtube.com → Create (camera icon) → Upload video → choose the .mp4.
2. Title: "Is South Carolina Heating Up? – Data Analysis Project". Paste the GitHub link in the description.
3. Choose "No, it's not made for kids", then set Visibility to Unlisted (anyone with the link can watch) unless your instructor wants Public.
4. Copy the video link and paste it into README.md ("Presentation video") and the report's title page.

Checklist before the live presentation

- Names, course, date and links filled in on slides 1 and 16, README and report.
- Practise once with a timer – aim for 14 minutes so you have a buffer.
- Have the PDF of the slides ready as a backup in case PowerPoint fails.
- Know your three numbers by heart: +2.9 °F since 1966, nights warm faster than days, +47% warm nights.